

Appendix - Hysteresis Test Results

Single PDF version containing all appendix tables extracted from the source document. Numerical values were preserved as provided in the appendix.

Table 1. Results obtained from hysteresis for loading of 2831N without infill

VB [0/90/0/90]s	n disc	Arrangement	Stiffness [kN/mm]	E load [J]	E unload [J]	E dissipated [J]	%E dissipated	displacement max [mm]
Pd2-0	2	()	1.37	3.03	2.58	0.45	14.74	2.79
Sd4-0	4	()()	0.74	5.40	4.76	0.65	11.95	4.35
Pd4-0	4	(())	2.76	1.72	1.38	0.33	19.31	1.67
Pd6-0	6	((()))	3.25	1.59	1.23	0.36	22.78	2.40
Sd6-0	6	()()()	0.51	8.10	7.17	0.93	11.53	6.38
Hd8-0	8	(())(())	1.50	3.26	2.71	0.55	16.91	3.60

Table 2. Results obtained from hysteresis for loading of 2681N without infill

CM [0/90/0/90]s	n disc	Arrangement	Stiffness [kN/mm]	E load [J]	E unload [J]	E dissipated [J]	%E dissipated	displacement max [mm]
Pd2-0	2	()	0.93	3.80	3.21	0.58	15.4	2.96
Sd4-0	4	()()	0.43	7.80	6.22	1.58	20.2	5.83
Pd4-0	4	((()))	2.01	2.09	1.67	0.41	19.9	2.22
Pd6-0	6	((()))	2.40	1.91	1.51	0.393	20.7	2.46
Sd6-0	6	()()	0.28	11.99	9.56	2.43	20.25	8.88
Hd8-0	8	((()))()	0.91	4.55	3.57	0.98	21.5	4.98

Table 3. Results obtained from hysteresis for loading of 2660N without infill

CM [90/0/-45/45]s	n disc	Arrangement	Stiffness [kN/mm]	E load [J]	E unload [J]	E dissipated [J]	% E dissipated	displacement max [mm]
Pd2-0	2	()	1.22	3.16	2.74	0.42	13.15	1.22
Sd4-0	4	()()	0.63	5.98	4.86	1.12	18.73	5.37
Pd4-0	4	((()))	1.72	2.30	1.86	0.44	19.21	2.64
Pd6-0	6	((()))	2.53	1.58	1.24	0.34	21.70	1.85
Sd6-0	6	()()()	0.64	6.63	5.39	1.24	18.76	6.13
Hd8-0	8	((()))()	1.14	3.61	2.98	0.63	17.55	4.08

Table 4. Results obtained from hysteresis for loading of 2831N with infill

VB [0/90/0/90]s	n disc	Arrangement	Infill thickness [mm]	Stiffness [kN/mm]	E load [J]	E unload [J]	E dissipated [J]	E dissipated [%]	displacement max [mm]
Pd2-1	2	(I)	1	1.42	2.69	1.67	1.01	37.71	3.93
Pd2-3	2	(I)	3	0.55	6.19	2.69	3.50	56.49	4.29
Pd2-5	2	(I)	5	0.24	11.60	3.70	7.90	68.09	7.38
Sd4-1	4	(I)(I)	1	0.69	4.63	3.01	1.62	34.99	4.23
Sd4-3	4	(I)(I)	3	0.41	10.30	5.43	4.87	47.30	7.31
Sd4-5	4	(I)(I)	5	0.16	19.52	7.88	11.63	59.61	14.38
Pd4-1	4	((I))	1	3.20	1.90	1.24	0.66	34.59	2.46
Pd4-3	4	((I))	3	1.37	3.50	1.99	1.50	43.03	3.20
Pd4-5	4	((I))	5	0.17	13.86	4.30	9.56	68.95	8.50
Pd6-1	6	((()))	1	2.98	2.24	1.52	0.72	32.09	2.74
Pd6-3	6	((()))	3	0.78	5.20	2.67	2.53	48.71	4.17
Pd6-5	6	((()))	5	0.23	12.26	4.54	7.72	62.94	8.33

Table 5. Results obtained from hysteresis for loading of 2681N with infill

CM [0/90/0/90]s	n disc	Arrangement	Infill thickness [mm]	Stiffness [kN/mm]	E load [J]	E unload [J]	E dissipated [J]	E dissipated [%]	displacement max [mm]
Pd2-1	2	(I)	1	1.93	2.21	1.28	0.93	42.08	1.67
Pd2-3	2	(I)	3	0.63	6.04	2.95	3.09	51.16	4.12
Pd2-5	2	(I)	5	0.22	10.97	3.44	7.53	68.64	7.50
Sd4-1	4	(I)(I)	1	0.80	5.16	3.34	1.82	35.29	4.40
Sd4-3	4	(I)(I)	3	0.33	11.83	6.60	5.23	44.19	8.83
Sd4-5	4	(I)(I)	5	0.32	23.72	9.91	13.82	58.24	18.20
Pd4-1	4	((I))	1	1.48	2.86	1.81	1.04	36.57	2.71
Pd4-3	4	((I))	3	1.01	5.22	2.28	2.94	56.32	3.82
Pd4-5	4	((I))	5	0.32	12.86	3.87	8.99	69.91	8.03
Pd6-1	6	((((I)))	1	2.52	2.12	1.46	0.66	31.18	2.29
Pd6-3	6	((((I)))	3	0.91	4.53	2.37	2.16	47.68	3.97
Pd6-5	6	((((I)))	5	0.37	10.91	3.61	7.30	66.91	8.23

Table 6. Results obtained from hysteresis for loading of 2660N with infill

CM [90/0/-45/45]s	n disc	Arrangement	Infill thickness [mm]	Stiffness [kN/mm]	E load [J]	E unload [J]	E dissipated [J]	E dissipated [%]	displacement max [mm]
Pd2-1	2	(I)	1	2.33	1.76	1.00	0.76	43.18	2.33
Pd2-3	2	(I)	3	0.79	4.72	2.14	2.58	54.66	0.79
Pd2-5	2	(I)	5	0.19	11.56	3.35	8.21	71.02	0.19
Sd4-1	4	(I)(I)	1	0.89	3.98	2.19	1.79	44.97	0.89
Sd4-3	4	(I)(I)	3	0.51	7.33	3.93	3.40	46.38	0.51
Sd4-5	4	(I)(I)	5	0.20	17.34	5.86	11.48	66.21	0.20
Pd4-1	4	((I))	1	2.99	2.00	1.32	0.68	34.00	2.99
Pd4-3	4	((I))	3	0.95	4.20	1.95	2.25	53.57	0.95
Pd4-5	4	((I))	5	0.49	10.03	3.03	7.00	69.79	0.49
Pd6-1	6	((((I)))	1	2.90	2.04	1.40	0.64	31.37	2.90
Pd6-3	6	((((I)))	3	1.08	3.89	2.24	1.65	42.42	1.08
Pd6-5	6	((((I)))	5	0.47	10.08	3.15	6.93	68.75	0.47